

Generation Interconnection System Impact Study Report

For Queue Project Q11-0040

James E. Rogers Energy Complex 138 kV

October 02, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0040. The Interconnection Customer proposes a 98.7 MW solar photovoltaic facility interconnecting at the James E. Rogers Energy Complex 138 kV substation in the Pennsylvania area. The request was received on May 27, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 35 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,245,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Carl Bailey - Otsego 138 kV	Norway Pt Hydropower Project - James E. Rogers Energy Complex 138 kV	112.7	277	6.0
Carl Bailey - Bon Aqua 138 kV	James E. Rogers Energy Complex - Norway Pt Hydropower Project 138 kV	108.7	279	6.0
Bon Aqua - Dry Gap 138 kV	Norway Pt Hydropower Project - James E. Rogers Energy Complex 138 kV	107.5	277	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Otsego - Carl Bailey 138 kV	Rebuild line with higher-capacity conductor	\$4,961,000	100% (\$4,961,000)
Carl Bailey - Bon Aqua 138 kV	Reconductor line	\$3,158,000	100% (\$3,158,000)
Bon Aqua - Dry Gap 138 kV	Reconductor line	\$2,126,000	100% (\$2,126,000)
	Total allocated to project		\$10,245,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.